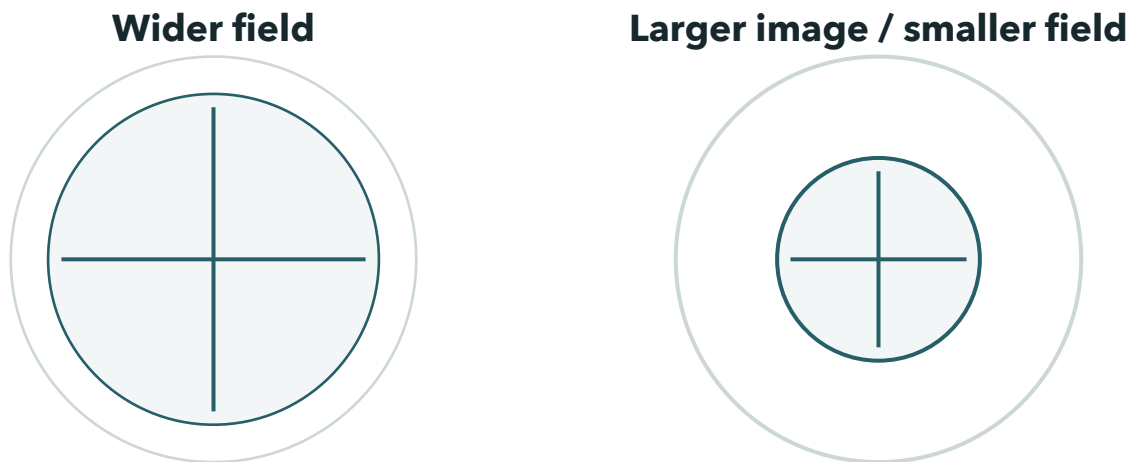


Imaging instruments and spatial resolution

Part 1 | An image can be larger but cover less retina

Condensing-lens power changes the field-magnification trade-off in indirect ophthalmoscopy.



Field, magnification and resolution are distinct properties

A wider sampled retinal field and a more magnified smaller field are shown as different instrument goals.

Use the instrument purpose

A higher-power condensing lens generally gives a wider retinal field and lower image magnification under comparable viewing conditions. Image orientation and magnification are separate properties; a magnification change does not by itself change the inversion mechanism.

Spatial and coherence sectioning differ

A confocal pinhole and OCT depth gating reject unwanted signals by different principles.

Technique

Feature	Selective principle	Main signal
Confocal	Spatial conjugate aperture	In-focus reflected light
OCT	Low-coherence interferometry	Depth-resolved backscatter
Conventional slit image	Illumination and viewing geometry	Optical section of visible tissue

Compare a redesign

Removing a confocal pinhole increases unwanted out-of-focus contribution; changing an OCT source’s coherence properties affects depth discrimination. Both are optical-sectioning systems, but an identical phrase does not make their physical gates interchangeable.

Resolution is anisotropic

Axial and lateral resolution can depend on different parameters.

OCT model

Feature	Direction	Dominant optical factor
Axial	Along the beam	Source bandwidth/coherence
Lateral	Across the beam	Focused spot/optics
Image sampling	Pixel or A-scan spacing	Must adequately sample the optical information

A design inference

Increasing display pixel count without changing optical resolution cannot reveal arbitrarily finer true structures. Conversely, an improved optical point-spread function can be wasted by inadequate sampling. Distinguish instrument resolution from image magnification and from the exported screenshot dimensions.

A vascular slab can inherit a false pattern

OCTA artefacts can resemble anatomy and need structural cross-checking.

Artefact

Feature	Appearance	Check
Projection	Superficial pattern repeated deeper	Compare overlying vessels
Segmentation error	Wrong tissue included in slab	Inspect structural B-scans
Shadowing	Reduced signal below opacity	Check attenuation and media

Interpretation

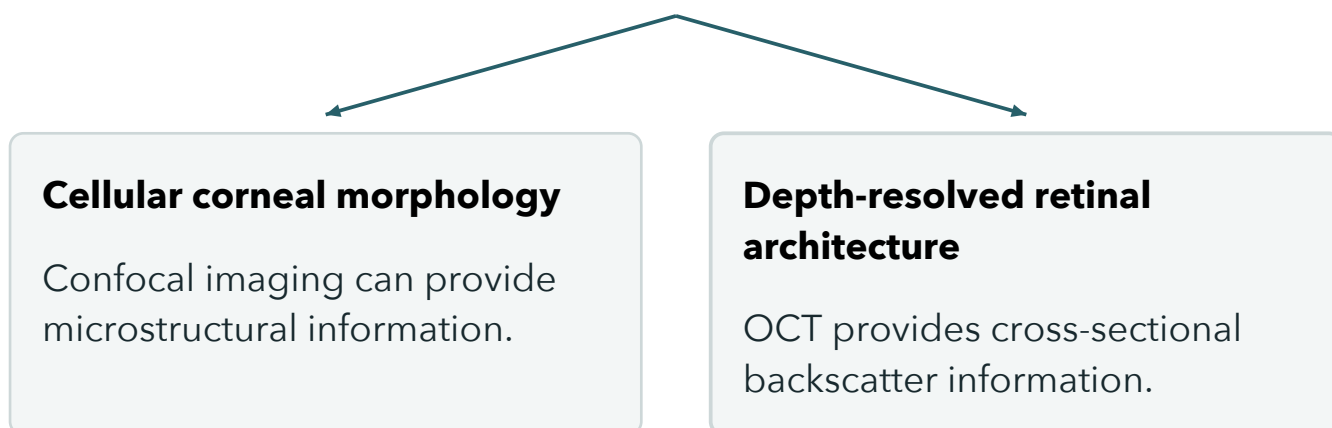
A deep-slab vessel pattern is stronger evidence when its position survives corrected segmentation and does not simply duplicate superficial vessels. Likewise, a low-signal region is not automatically proof of absent perfusion. Read the flow map alongside the structural image.

Match the instrument to the question

Each technique preserves different information and has different access limitations.

What must be resolved?

Choose the target before the machine



Three checks

Is the optical path clear enough?

Is the structure within the sampled field and depth?

Does the displayed boundary reflect anatomy or processing?